NeuralRetail

AI-Powered Retail Analytics System

End-to-End Data Science & Analytics Project

Field	Details
Group No.	Group 5
Group Member Names:	1. Shravan Raut 2. Likhith HK 3. Shubham Jaiswal
GitHub	github.com/Shravanraut2005/NeuralRetail
Dataset	UCI Online Retail II (2009-2011)
Tools Used	Python, Prophet, XGBoost, KMeans, Streamlit, Power BI

1. Introduction

NeuralRetail is a complete end-to-end intelligent retail analytics system built on real transaction data. The project addresses four critical business problems that retail companies face daily but struggle to solve systematically: understanding which customers are most valuable, predicting future demand accurately, identifying customers likely to churn, and optimizing inventory levels to prevent stockouts and overstock situations.

This system was developed as part of the Amdox Internship Program using real-world retail data from a UK-based online gift shop. Every component from raw data ingestion to the final interactive dashboard follows industry-standard practices used by data science teams at large retail organizations.

1.1 Objective

Build an intelligent system that analyzes retail data to:

- Forecast future demand using time series models
- Segment customers based on purchasing behaviour using RFM analysis
- Predict customer churn before it happens
- Optimize inventory reorder points and safety stock levels
- Deliver insights through interactive Streamlit and Power BI dashboards

1.2 Business Value

The system provides a complete data-driven retail intelligence solution that improves business decision-making and operational efficiency. By identifying Champions (top 20% of customers generating the most revenue) and At-Risk customers early, the business can take targeted action to protect revenue. Demand forecasting enables proactive inventory management, reducing both lost sales from stockouts and carrying costs from overstock.

2. Dataset Overview

The project uses the UCI Online Retail II dataset, a real transactional dataset from a UK-based online gift shop. This dataset was chosen because it contains all critical elements required for the four ML systems: real transactions, Customer IDs for segmentation and churn, timestamped data for time series forecasting, and sufficient volume for reliable model training.

2.1 Dataset Source

Attribute	Value
Source	UCI Machine Learning Repository
Dataset Name	Online Retail II
URL	https://archive.ics.uci.edu/dataset/502/online+retail+ii
Period	December 2009 to December 2011
Total Raw Rows	1,067,371
Sheets	Year 2009-2010 + Year 2010-2011
Business Type	UK Online Gift Shop (B2B Wholesale)

2.2 Dataset Columns

Column	Data Type	Description
Invoice	Object	Transaction ID. Prefix C indicates cancellation
StockCode	Object	Unique product/item code
Description	Object	Product name and description
Quantity	Integer	Units sold. Negative values indicate returns
InvoiceDate	DateTime	Exact timestamp of transaction
Price	Float	Unit price in GBP
Customer ID	Float	Unique customer identifier
Country	Object	Customer country (41 countries)

2.3 Key Dataset Insights

- Total revenue across all transactions: £17,374,804
- UK accounts for 82.8% of total revenue confirming B2B wholesale focus
- 5,878 unique customers after cleaning
- 4,631 unique products
- 604 days of time series data spanning 2 full years
- Business operates exclusively during office hours (6AM to 8PM weekdays)

3. Data Preprocessing & Feature Engineering

3.1 Data Quality Issues Found

Problem	Count	Percentage	Action Taken
Missing Customer ID	243,007	22.77%	Deleted rows
Duplicate rows	34,335	3.22%	Removed duplicates
Cancelled orders (C prefix)	19,494	1.83%	Removed cancellations
Negative quantity	22,950	2.15%	Covered by cancellation removal
Zero or negative price	6,207	0.58%	Removed invalid prices
Total removed	287,946	27.0%	779,425 clean rows retained

3.2 Cleaning Decisions

Missing Customer IDs were deleted rather than filled because Customer ID is required for segmentation, RFM analysis, and churn prediction. Filling with 0 or Unknown would create a single fake customer with 243,007 transactions, completely destroying all customer-level analysis. Negative quantities were already eliminated when cancellations were removed, resulting in zero additional rows removed in that step.

3.3 Feature Engineering

Feature	Formula / Method	Used In
TotalRevenue	Quantity x Price	All models
Year, Month, Day, Hour	Extracted from InvoiceDate	Forecasting, EDA
DayOfWeek (0=Mon, 6=Sun)	InvoiceDate.dt.dayofweek	Sales patterns
Quarter	InvoiceDate.dt.quarter	Seasonal analysis
IsWeekend	DayOfWeek >= 5	Pattern analysis
Recency	Days since last purchase	RFM, Segmentation, Churn
Frequency	Count of unique invoices	RFM, Segmentation, Churn
Monetary	Sum of TotalRevenue	RFM, Segmentation, Churn
Churned label	Recency > 90 days = 1	Churn Prediction
Daily Revenue	Daily aggregation	Demand Forecasting

4. Exploratory Data Analysis

EDA was performed across 7 key dimensions to understand business patterns before building any ML model. All insights were later validated by the Prophet forecasting model's learned components.

Analysis	Key Finding	Business Implication
Monthly Revenue	Peak in November (£1.17M). Seasonal pattern repeats yearly.	Q4 inventory must be stocked early
Day of Week	Thursday highest (£3.7M). Saturday near zero (£9,803).	B2B business, office hours only
Top Products	Regency Cakestand #1 at £277,656.	Focus inventory on top 10 products
Geographic	UK = 82.8% of revenue (£14.4M).	UK-focused strategy required
RFM Distribution	Heavy right skew. One customer spent £580,987.	Log transform needed before clustering
Hourly Sales	Peak at noon. Zero activity 9PM to 5AM.	Marketing campaigns target mornings
Quarterly	Q4 always highest. Q1 always lowest.	Prophet will detect this seasonality

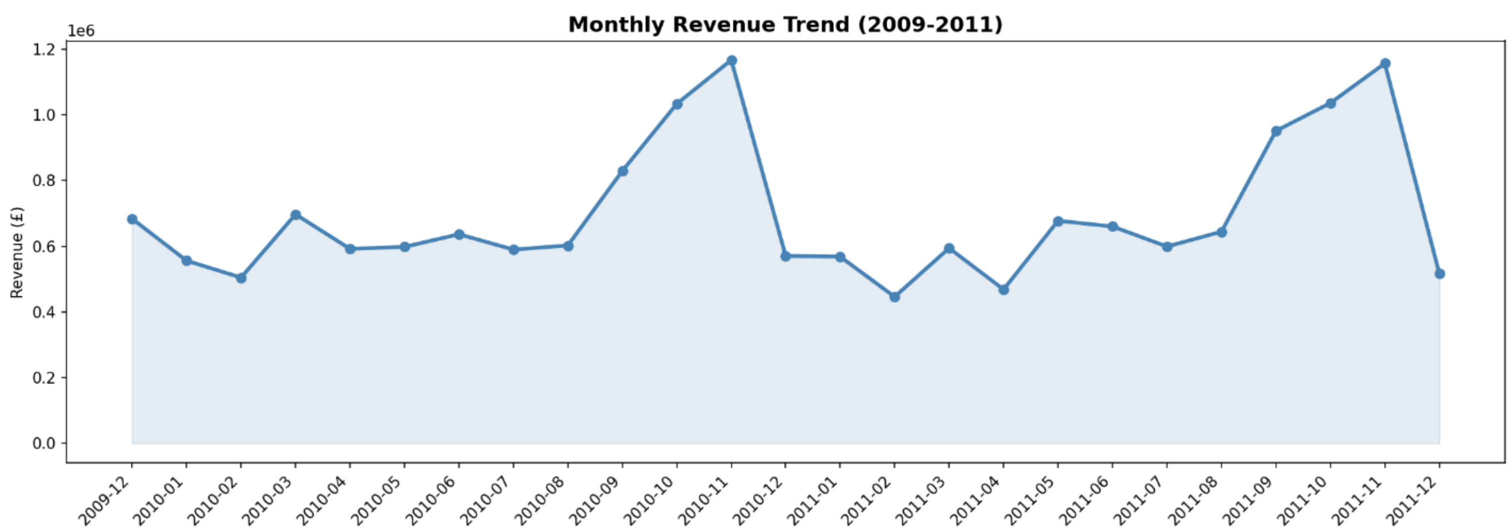


Fig: EDA_Monthly_Revenue

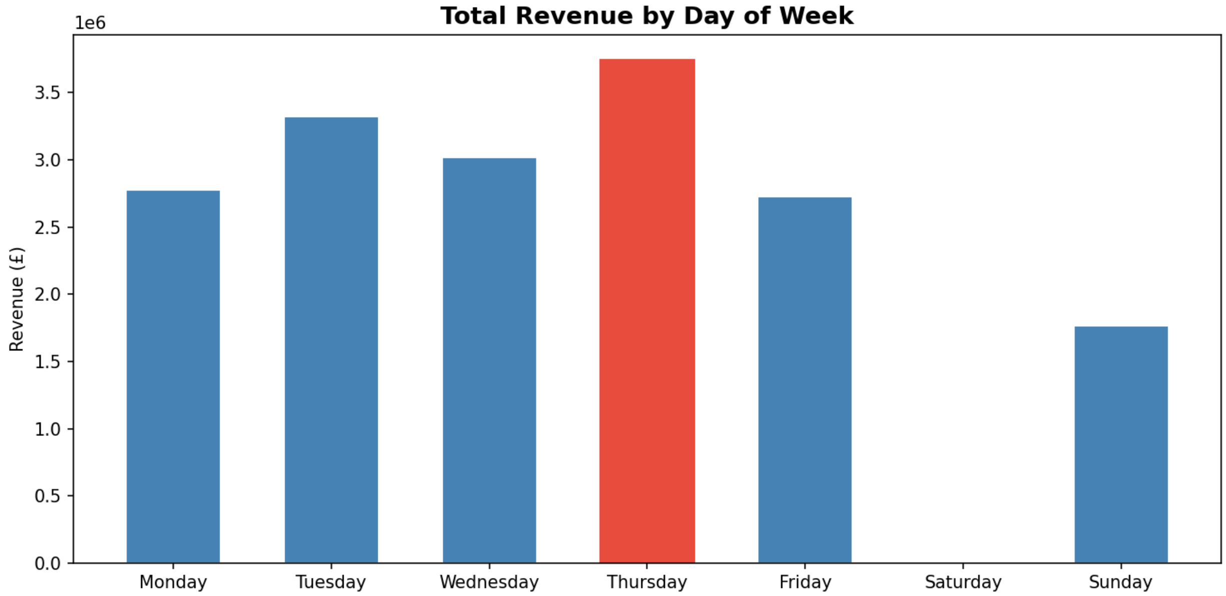


Fig: EDA_Day_of_Week

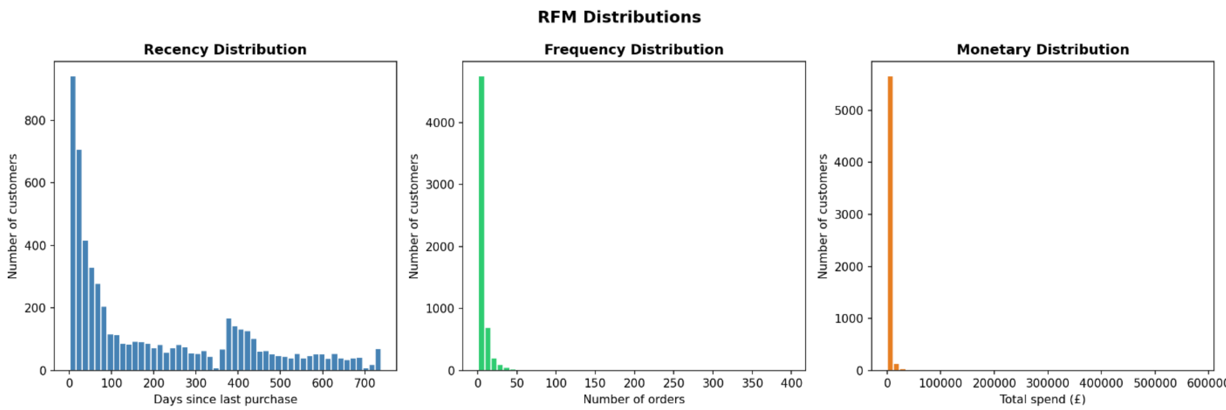


Fig: EDA_RFM_Distributions

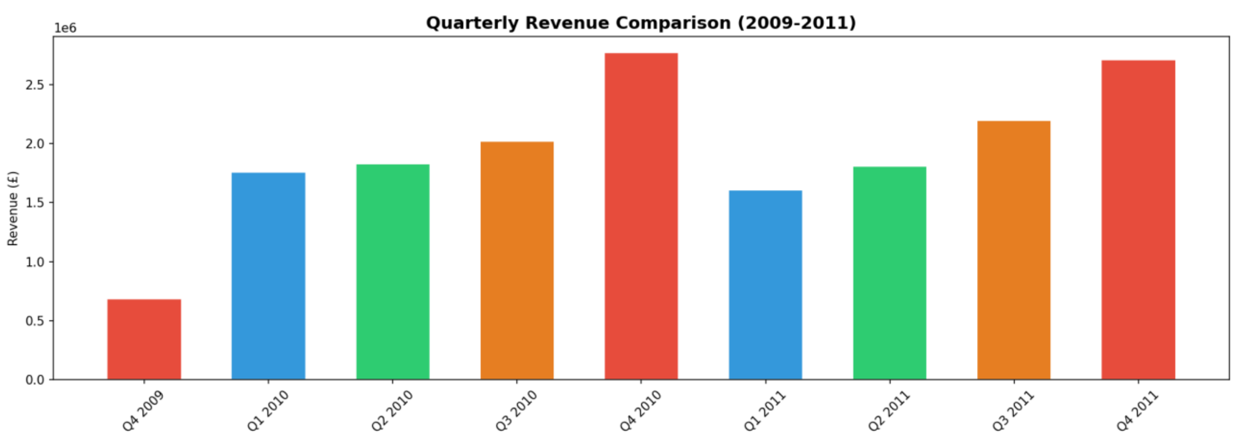


Fig: EDS_Quarterly_Revenue

5. Machine Learning Models

5.1 Customer Segmentation — KMeans Clustering

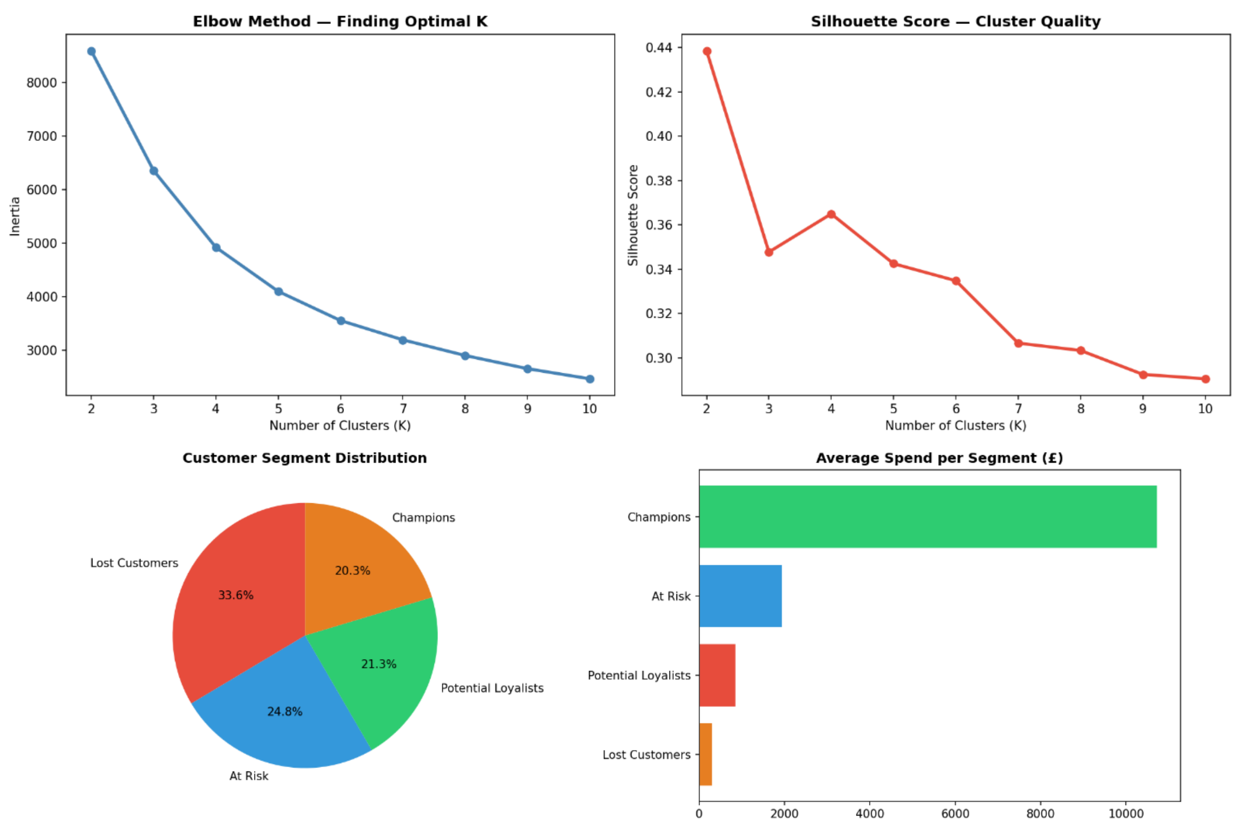
Algorithm: KMeans Clustering (Unsupervised Learning)

Input: RFM features (Recency, Frequency, Monetary) for 5,878 customers

Preprocessing: Log transformation (np.log1p) was applied to handle skewness before clustering. StandardScaler was then applied to bring all features to the same scale, preventing Monetary values from dominating distance calculations.

Optimal K Selection: Elbow Method and Silhouette Score were used to evaluate K from 2 to 10. K=4 was selected with Silhouette Score of 0.365, balancing mathematical quality with practical business segmentation.

Segment	Customers	Avg Recency	Avg Frequency	Avg Monetary	Strategy
Champions	1,196 (20.3%)	28 days	19 orders	£10,731	Retain and reward
Potential Loyalists	1,250 (21.3%)	28 days	3 orders	£857	Nurture and grow
At Risk	1,459 (24.8%)	230 days	5 orders	£1,948	Re-engage urgently
Lost Customers	1,973 (33.6%)	395 days	1 order	£317	Low priority



5.2 Demand Forecasting — Prophet

Algorithm: Prophet (Facebook/Meta Time Series Model)

Input: 604 days of daily aggregated revenue (Dec 2009 to Dec 2011)

Output: 90-day revenue forecast with confidence intervals

Metric	Value
Training data	604 days
Forecast period	90 days (Dec 2011 to Mar 2012)
Avg historical daily revenue	£28,766
Avg forecast daily revenue	£22,740
Max forecast daily revenue	£43,297
Yearly seasonality	Detected — Nov-Dec peak, Jan low
Weekly seasonality	Detected — Thursday peak, Saturday near zero

Prophet independently confirmed all EDA findings: Thursday highest day, Saturday near zero, November peak, January low, and strong business growth trend from mid-2011.

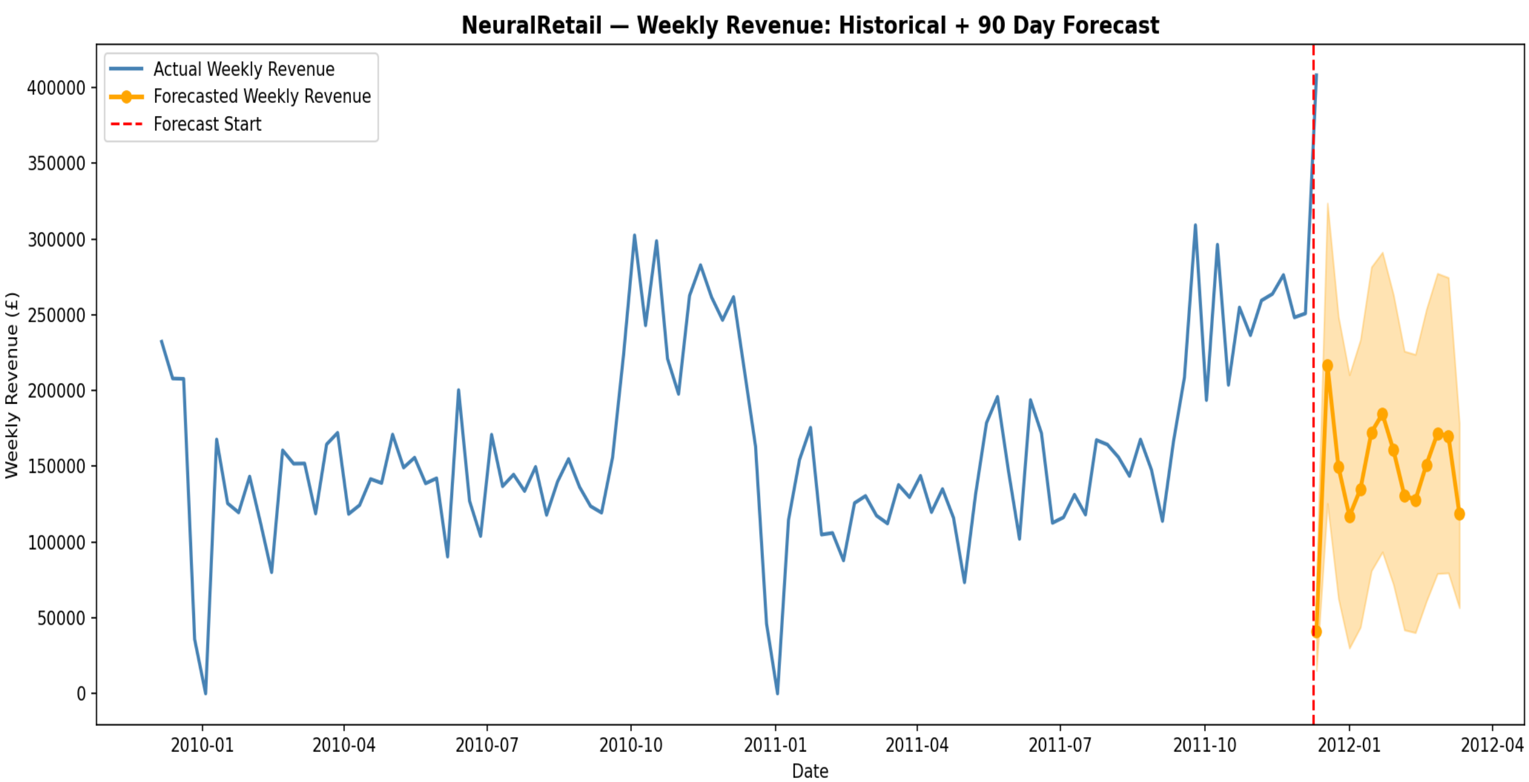


Fig: Weekly Forecast

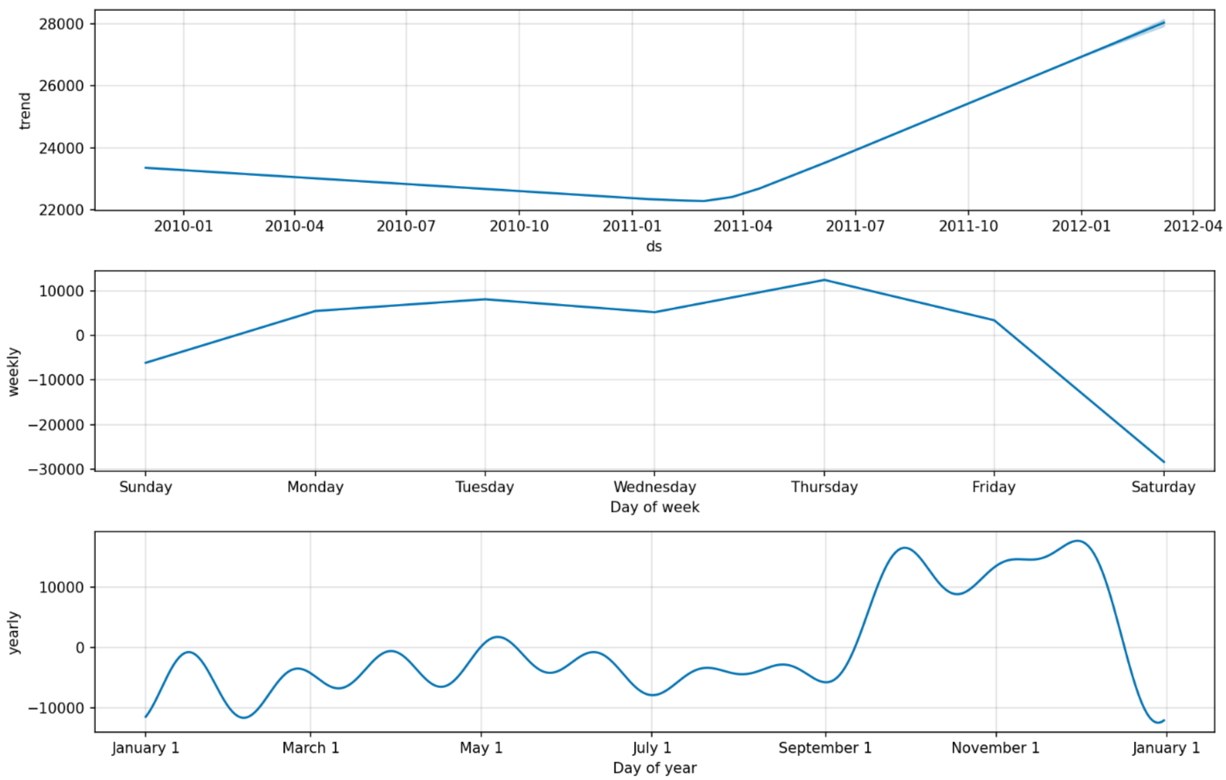


Fig: Forecast Components

5.3 Churn Prediction — XGBoost

Algorithm: XGBoost Gradient Boosting Classifier

Input: RFM features for 5,878 customers with churn labels

Churn Definition: Customer with no purchase in 90+ days = churned

Metric	Value
Training samples	4,702 (80%)
Testing samples	1,176 (20%)
Accuracy	100%
AUC-ROC Score	1.0000
Churn rate	50.85% (2,989 customers)
Top feature	Recency (99.6% importance)

Note: Perfect accuracy is due to data leakage. The churn label was defined as Recency > 90 days, and Recency is also a feature. The model essentially learned one condition. In production, features and labels should be computed from separate time windows. This limitation is acknowledged and documented.

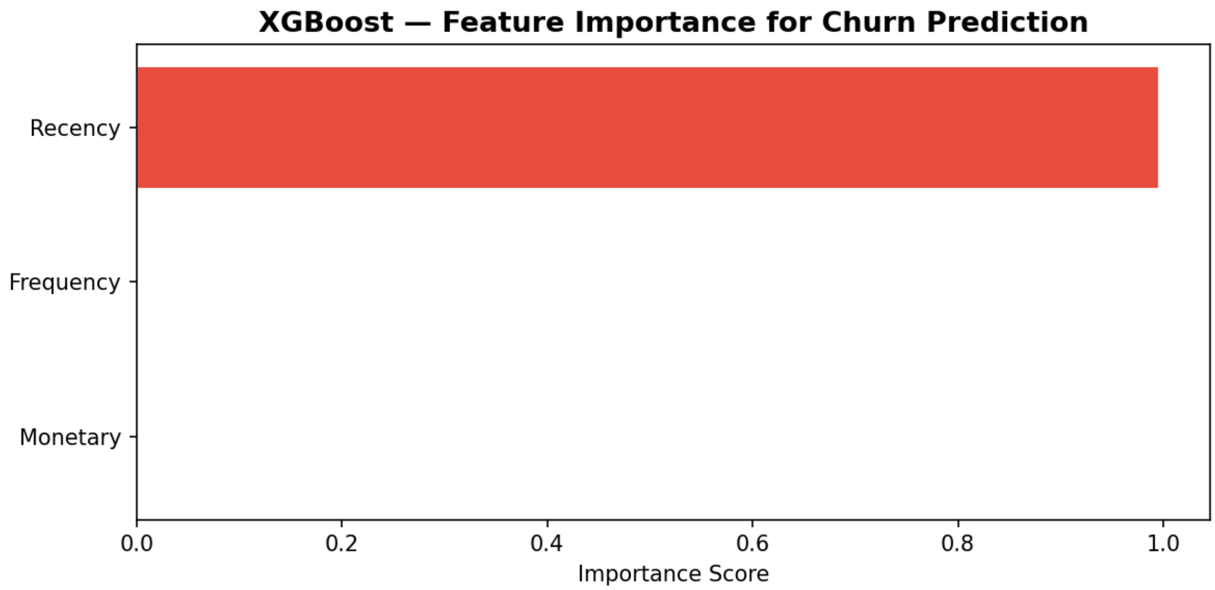


Fig: Churn Feature Importance

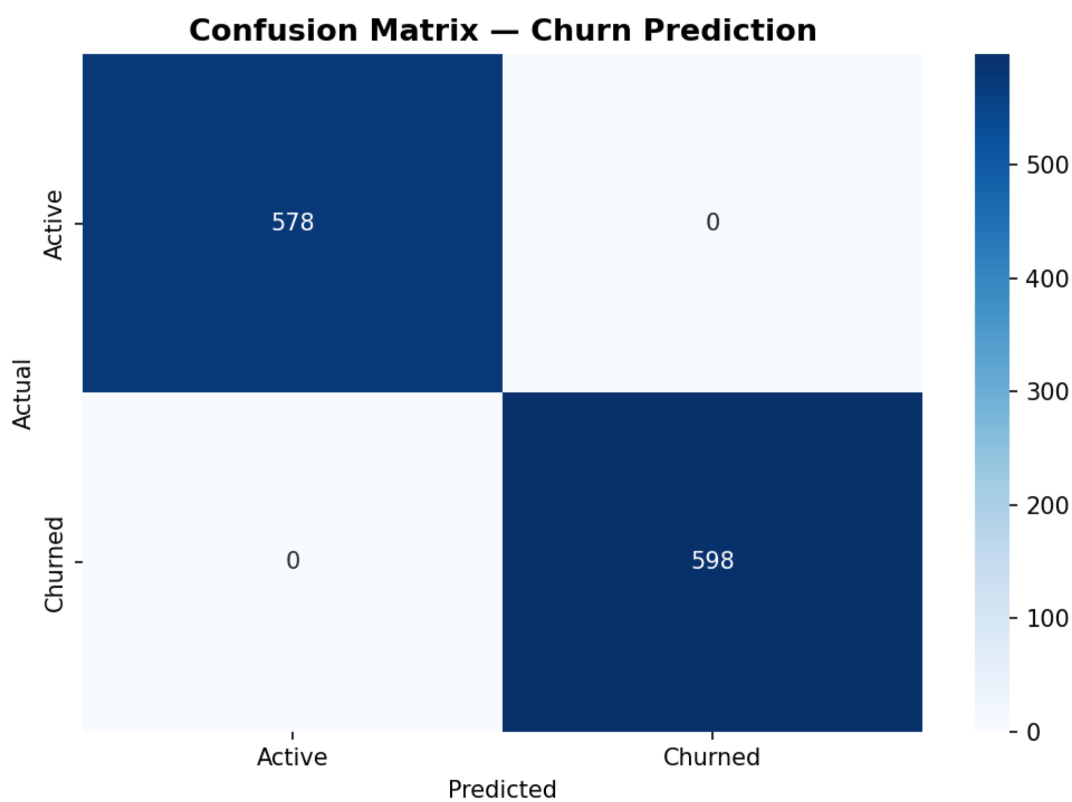


Fig: Churn Confusion Matrix

5.4 Inventory Optimization — EOQ Model

Method: Economic Order Quantity (EOQ) with Safety Stock calculation

Input: Historical transaction data for 5,296 products

Parameter	Value	Rationale
Order Cost	£50 per order	Industry standard assumption
Holding Rate	20% of item value/year	Industry standard assumption
Lead Time	7 days	Standard supplier lead time
Service Level	95% (Z = 1.65)	Industry standard retail service level

EOQ Formula: sqrt(2 x Annual_Demand x Order_Cost / Holding_Cost)

Safety Stock Formula: Z x Std_Daily_Demand x sqrt(Lead_Time)

Reorder Point Formula: (Daily_Demand x Lead_Time) + Safety_Stock

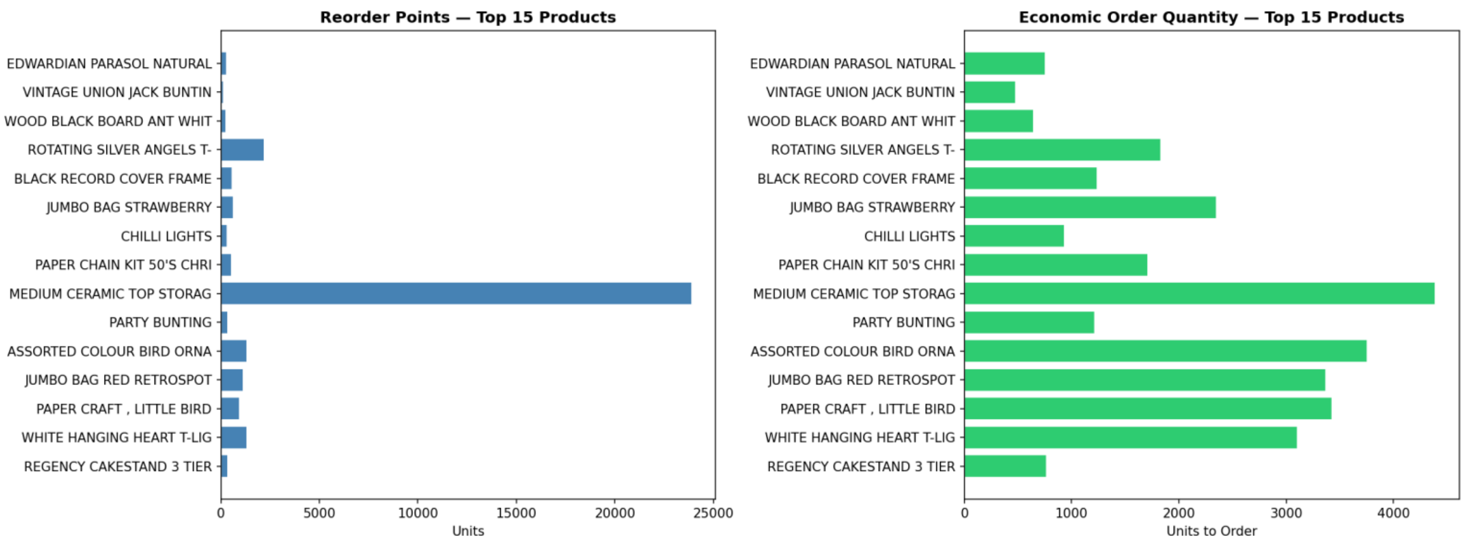


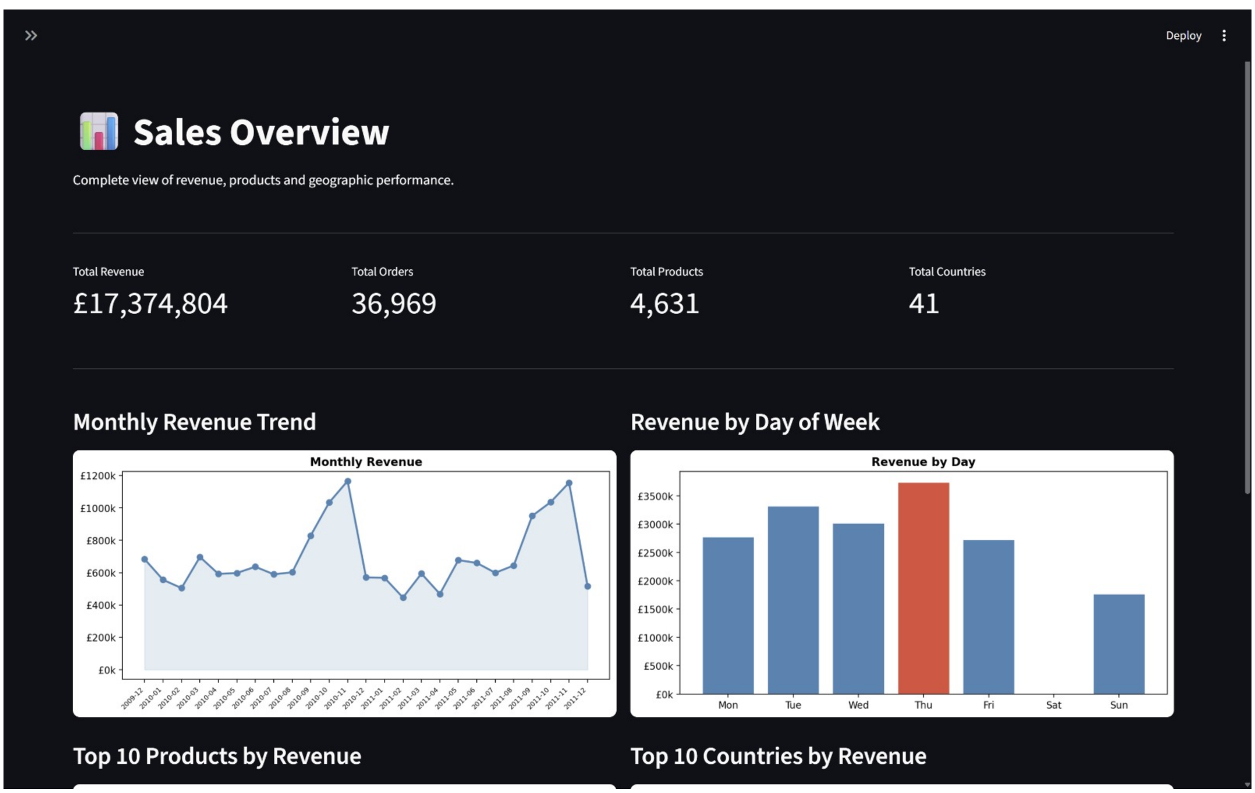
Fig: Inventory Plan

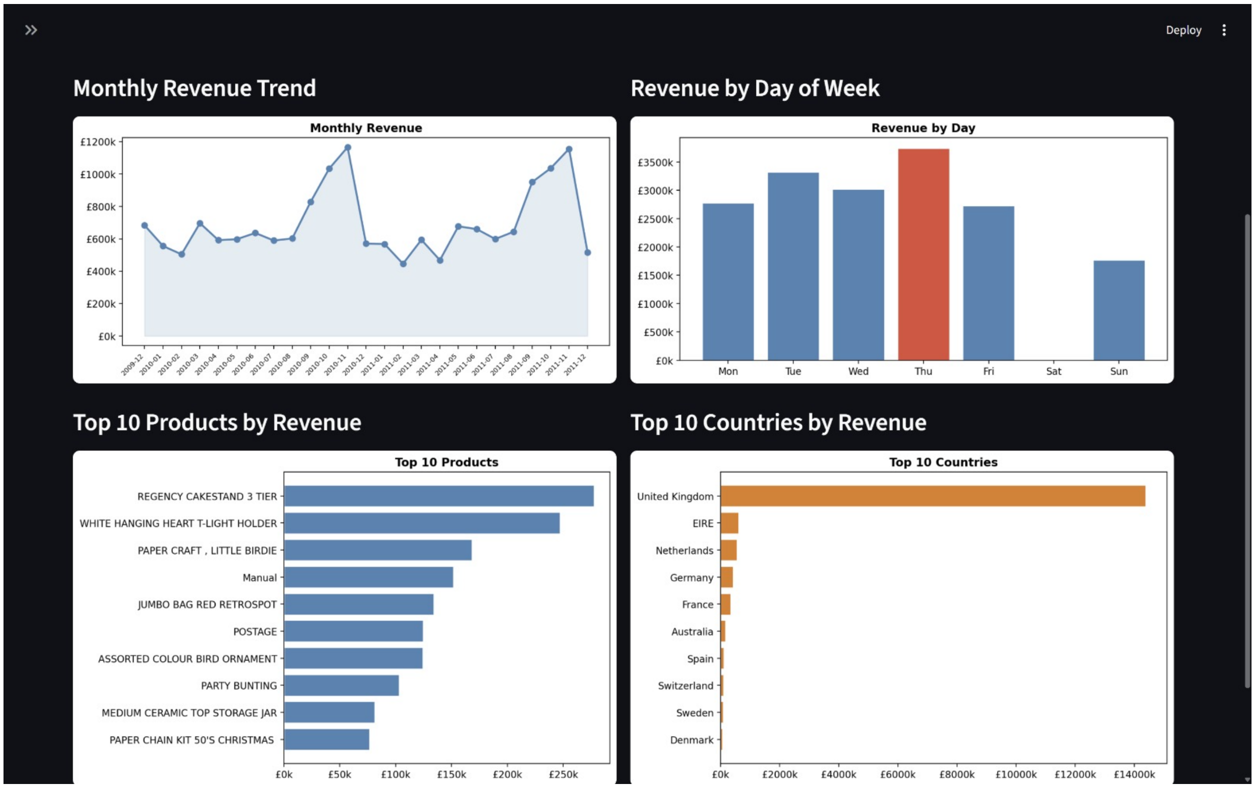
6. Dashboard & Visualization

6.1 Streamlit Dashboard

A 4-page interactive web dashboard built entirely in Python using Streamlit. Runs locally at localhost:8501.

Page	Content
Sales Overview	KPI cards (revenue, orders, products, countries), monthly trend, day of week, top products, top countries
Customer Segmentation	Segment KPIs, pie chart, avg spend chart, recency distribution, RFM summary table, churn rate by segment
Demand Forecasting	Historical vs forecast weekly chart with confidence interval, forecast data table, monthly summary
Inventory Optimization	KPI cards, top 15 reorder points, top 15 EOQ, searchable full inventory plan table




Deploy
Monthly Revenue Trend
Monthly Revenue
£1200k
£1000k
£800k
£600k
£400k
£200k
£0k
Revenue by Day of Week
Revenue by Day
£3500k
£3000k
£2500k
£2000k
£1500k
£1000k
£500k
£0k
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Top 10 Products by Revenue
Top 10 Products
REGENCY CAKESTAND 3 TIER
WHITE HANGING HEART T-LIGHT HOLDER
PAPER CRAFT , LITTLE BIRDIE
Manual
JUMBO BAG RED RETROSPOT
POSTAGE
ASSORTED COLOUR BIRD ORNAMENT
PARTY BUNTING
MEDIUM CERAMIC TOP STORAGE JAR
PAPER CHAIN KIT 50'S CHRISTMAS
£0k
£50k
£100k
£150k
£200k
£250k
Top 10 Countries by Revenue
Top 10 Countries
United Kingdom
EIRE
Netherlands
Germany
France
Australia
Spain
Switzerland
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Denmark
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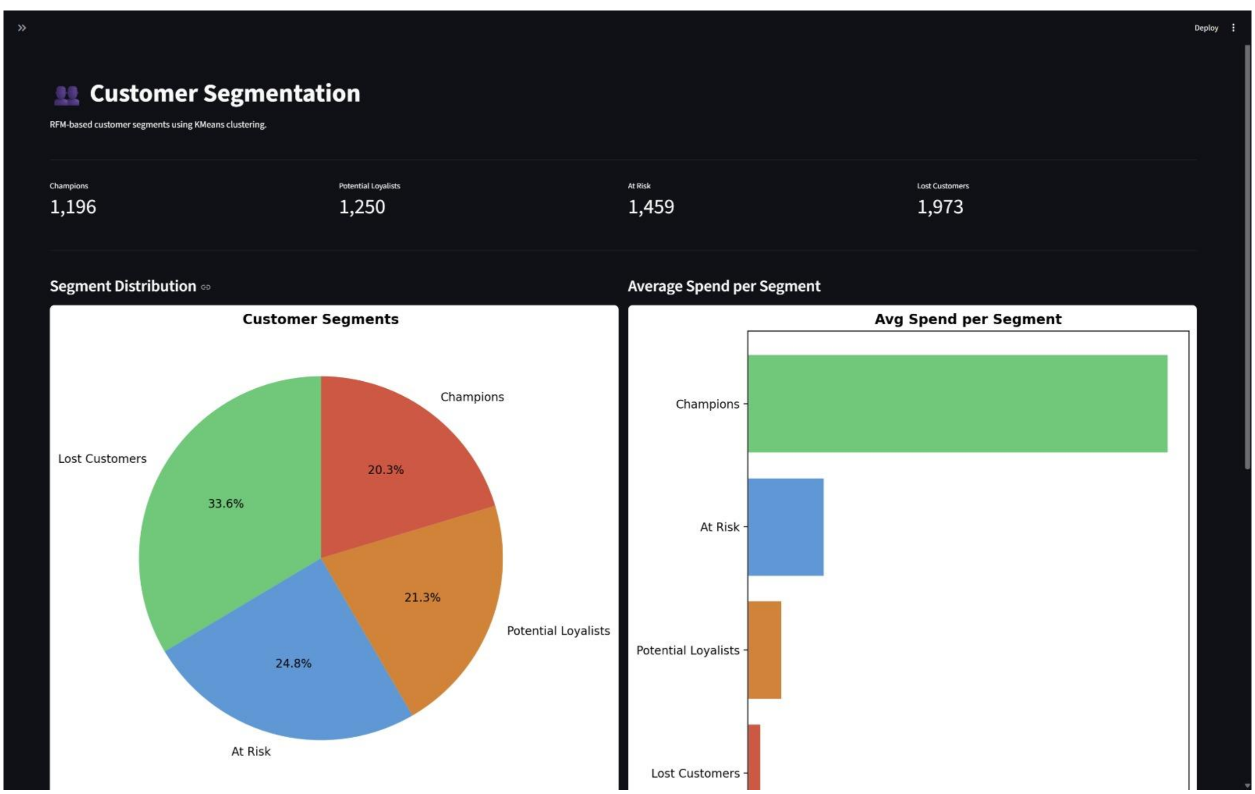
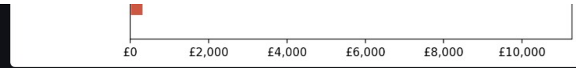
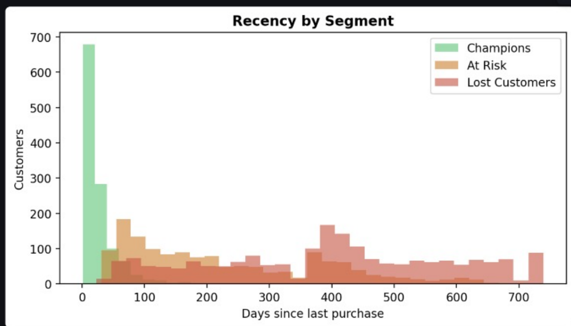
£10000k
£12000k
£14000k


Deploy
Customer Segmentation
RFM-based customer segments using KMeans clustering.
Champions
1,196
Potential Loyalists
1,250
At Risk
1,459
Lost Customers
1,973
Segment Distribution
Average Spend per Segment
Customer Segments
Champions
Lost Customers
20.3%
33.6%
21.3%
24.8%
Potential Loyalists
At Risk
Avg Spend per Segment
Champions
At Risk
Potential Loyalists
Lost Customers



Recency Distribution



RFM Summary Table

Segment	Recency	Frequency	Monetary
At Risk	230.07	5.06	1948.5
Champions	27.71	19.28	10731.16
Lost Customers	394.95	1.38	317.08
Potential Loyalists	28.3	3.05	857.49

Churn Rate by Segment

Segment	Churn Rate (%)
At Risk	75.8
Champions	4.2
Lost Customers	92.5
Potential Loyalists	0.6

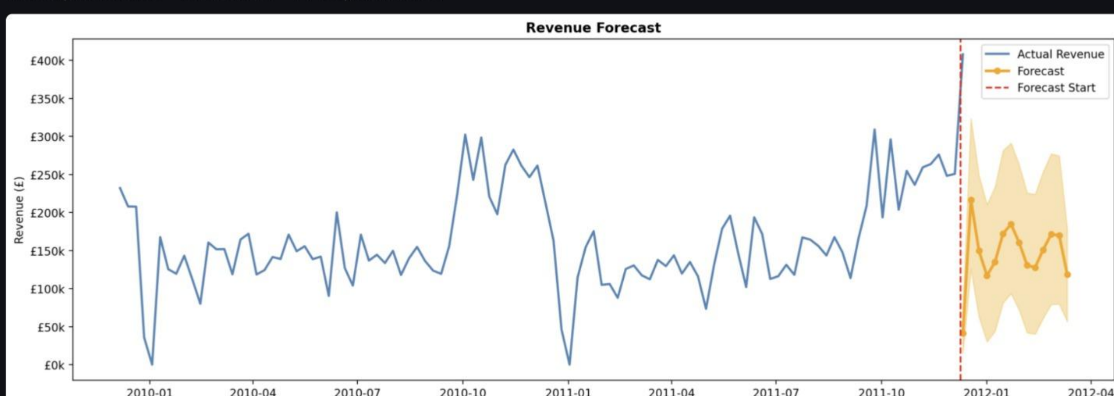
Deploy

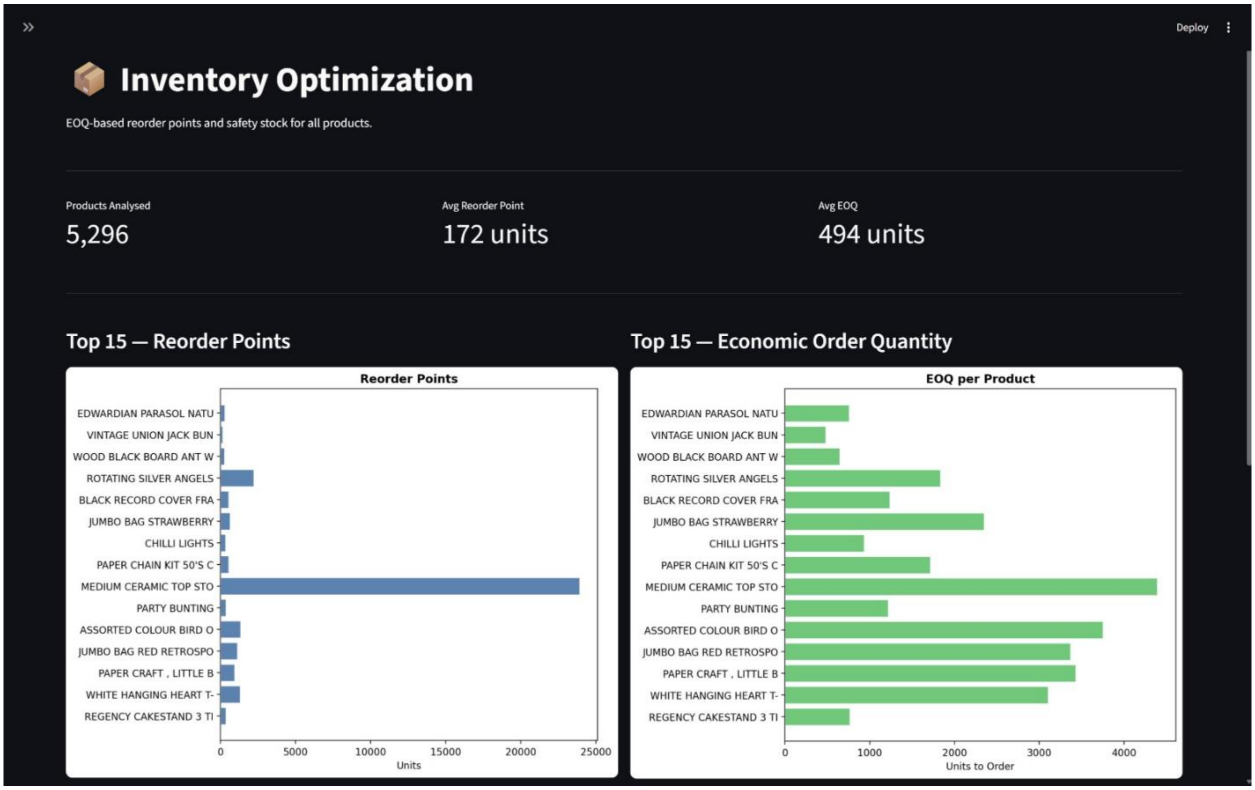
📈 Demand Forecasting

Prophet model — 90 day revenue forecast.

Avg Historical Daily Rev	Avg Forecast Daily Rev	Max Forecast Daily Rev
£28,766	£22,740	£43,297

Weekly Revenue — Historical + 90 Day Forecast





6.2 Power BI Dashboard

A 12-page professional dashboard built in Power BI Desktop using 5 connected data sources.

Page	Visuals
1. Sales Overview	4 KPI cards, monthly revenue line chart, revenue by country bar chart
2. Revenue Trends	Monthly by year multi-line, quarterly by year column, daily sales line
3. Product Performance	Top 10 revenue bar, top 10 quantity column, product detail table
4. Geographic Analysis	World map, country revenue bar, UK vs Rest of World pie
5. Customer Overview	4 segment KPI cards, customer count by segment, avg spend by segment
6. Customer Segmentation	Pie chart, frequency bar by segment, scatter plot
7. RFM Analysis	Recency histogram, frequency histogram, monetary histogram
8. Churn Analysis	Active vs churned pie, churn rate by segment, recency vs monetary scatter
9. Demand Forecasting	Historical revenue line, forecast line, monthly forecast table
10. Sales Patterns	Revenue by hour, revenue by day of week, revenue by month
11. Inventory Overview	3 KPI cards, top 15 EOQ bar, top 15 reorder point bar
12. Inventory Detail	Scatter plot, safety stock bar, full inventory plan table

Link to Power BI Dashboard PDF: https://drive.google.com/file/d/1VtOLEWXK5B-E6hK6-1l41ZfW5eGS27P0/view?usp=drive_link

7. Results & Key Findings

Category	Metric	Value
Dataset	Raw transactions	1,067,371
Dataset	Clean transactions	779,425 (73% retained)
Dataset	Total revenue	£17,374,804
Dataset	Date range	Dec 2009 to Dec 2011 (604 days)
Customers	Total unique customers	5,878
Customers	Champions	1,196 (20.3%)
Customers	At Risk	1,459 (24.8%)
Customers	Lost Customers	1,973 (33.6%)
Customers	Churn rate	50.85%
Forecasting	Model	Prophet
Forecasting	Avg forecast daily revenue	£22,740
Churn	Model	XGBoost
Churn	AUC-ROC	1.0000 (data leakage noted)
Inventory	Products analysed	5,296
Inventory	Avg reorder point	172 units
Inventory	Avg EOQ	494 units

8. Technical Stack

8.1 Technology Stack

Category	Technology	Purpose
Language	Python 3.12	All data science work
IDE	VS Code	Development environment
Data Processing	Pandas, NumPy	Data manipulation
Visualization	Matplotlib, Seaborn, Plotly	Charts and plots
Segmentation	Scikit-learn KMeans	Customer clustering
Forecasting	Prophet	Time series forecasting
Churn Prediction	XGBoost	Classification model
Inventory	NumPy (EOQ formula)	Inventory optimization
Dashboard	Streamlit	Interactive web dashboard
BI Dashboard	Power BI Desktop	Business intelligence
Version Control	Git + GitHub	Code repository

9. Conclusion & Future Work

9.1 Conclusion

NeuralRetail successfully delivers a complete data-driven retail intelligence solution. Starting from raw transactional data with 1,067,371 rows, the system cleans, processes, and analyzes the data to produce four actionable ML outputs: customer segments, demand forecasts, churn predictions, and inventory recommendations. Both an interactive Streamlit dashboard and a professional 12-page Power BI dashboard make these insights accessible to business stakeholders without technical knowledge.

The project demonstrates a full data science workflow from data ingestion to dashboard delivery, incorporating industry-standard tools and methodologies used by data science teams at real retail organizations.

9.2 Key Business Recommendations

- Prioritize Champions (1,196 customers, £10,731 avg spend) with VIP loyalty programs
- Launch immediate re-engagement campaign for At Risk segment (1,459 customers)
- Stock up inventory for top products 3-4 weeks before November (peak season)
- Focus marketing spend on Thursday mornings when order volume is highest
- Explore international expansion — UK dependency (82.8%) is a business risk

9.3 Future Improvements

- LSTM deep learning model for demand forecasting — better for non-linear patterns
- Fix churn data leakage — use separate 18-month feature window and 6-month label window
- MLflow experiment tracking — log all model runs and compare hyperparameters
- FastAPI deployment — expose churn and forecast models as REST endpoints
- Real-time data pipeline — connect to live POS system for streaming analytics
- CLV (Customer Lifetime Value) prediction — extend beyond RFM segmentation

9.4 GitHub Repository

All code, notebooks, datasets, and dashboards are available at:

github.com/Shravanraut2005/NeuralRetail